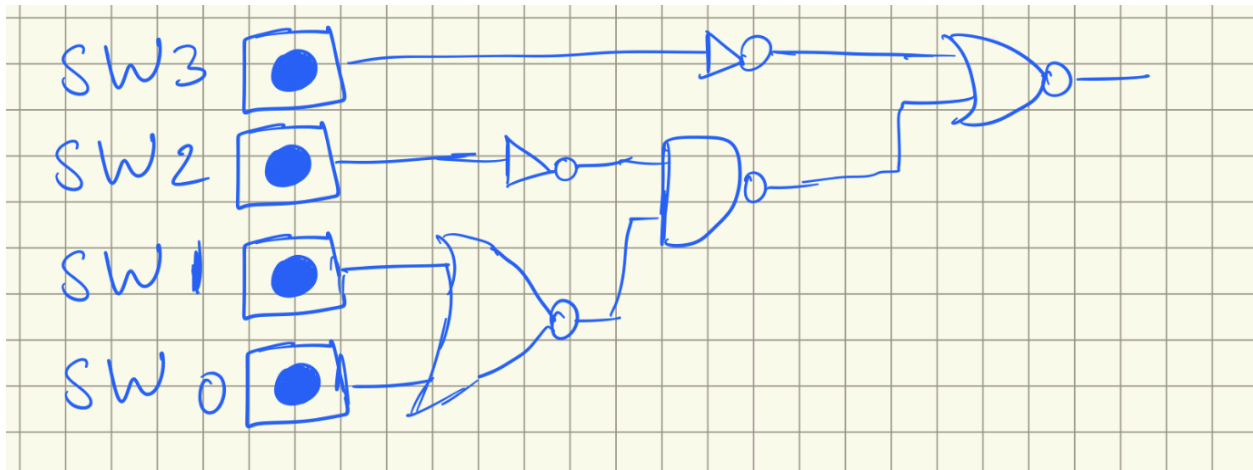


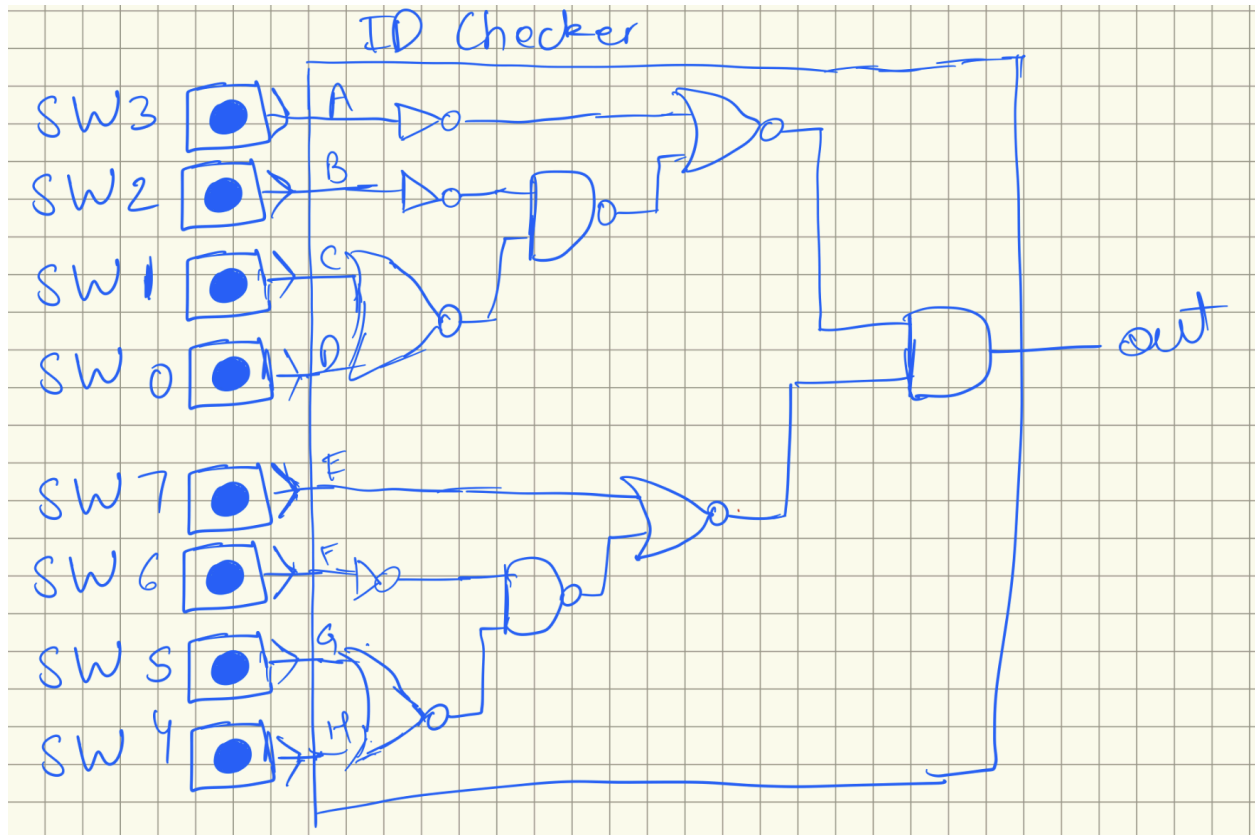
Task 1



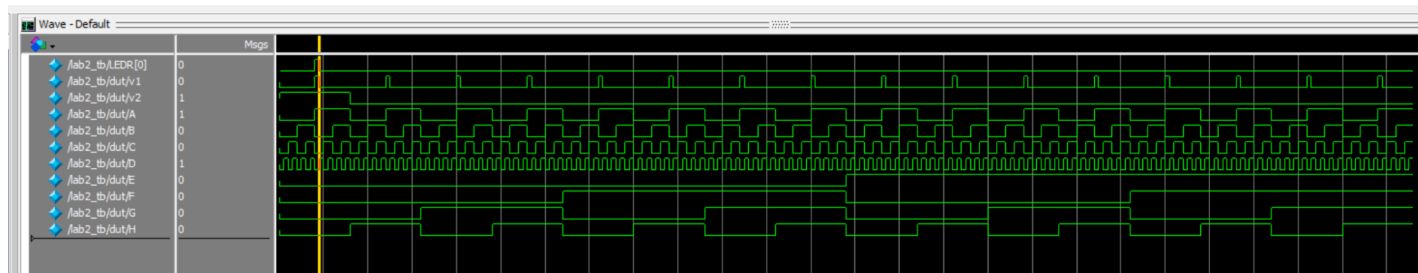
Explanation: The last digit of my student number is 8, which is 1000 according to the given table. So, logically, in order to recognize 8, only switch 3 shall be on and the rest of the switches off to be correct. This circuit only outputs a 1 (on state) when switch 3 is on and the rest of the switches are off.

Task 2:

My Module:

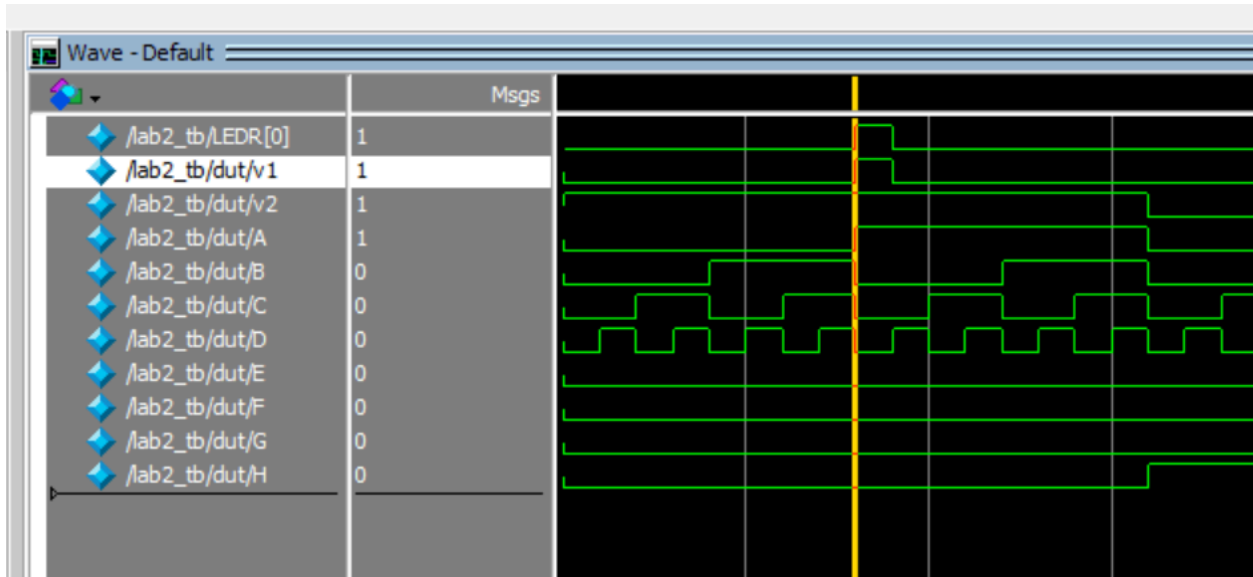


Wave forms



This screenshot shows the entire circuit simulation. Signals A to H are the input signals from the respective switches as labelled in the ID_CHECKER module. v1 signal is the output of the circuit that checks SW0 to SW3 and v2 signal is the output for the other circuit.

As we can see in the waveform, LEDR[0] only is 1 once. If we zoom in,



Here, we see that LEDR[0] is 1 when v1 and v2 and signal A are 1. Here, all the other signals are zero and only the A input and v1 and v2 intermediate signals (which represent output of each checker) are active. This module, according to the waveform, only outputs a 1 (on) only on the input signal where SW3, which corresponds to A, is on as well.

Thus, we have shown this module correctly checks for the last 2 digits of my student ID, which are 08. In binary: 0000 1000

Time spent: 5 hours